

NEW BATTERY TECHNOLOGIES

The electronics industry has been working towards the twin goals of reducing the size of devices and making them work for a longer time. Both goals are being achieved today mainly because of advances in IC technology – today's chips are not only very small but are also designed to consume very little power, by reducing energy leakage, etc. As a result, the devices they are embedded in also have smaller size and a longer battery life.

Battery technology, on the other hand, has not kept pace with this development – batteries have neither become significantly small nor has there been any significant improvement in their capacity. This has been a much-discussed topic over the past few years.

The trend seems to be shifting, and we now hear of a lot of research and advancements in battery technologies. Perhaps over the next year, some of these might get incorporated into products, to result in cellphones, laptops and other devices with a much longer battery life and smaller footprint than the current ones.

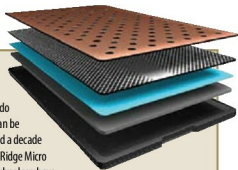
One notable development is the use of thin-film technology to make power sources, especially suitable for applications such as smart cards and sensors. These batteries can be flexible if fabricated on thin plastics. They charge fast and offer good energy and power densities in a small form factor. They also demonstrate very low equivalent series resistance (ESR).

They are also tolerant to a wide range of temperatures. Since they do not self-discharge, they can be used for a period of around a decade or so. Companies like Oak Ridge Micro Energy and Front Edge Technology have commercialised thin-film power sources.

Then, there is a Swiss company called ReVolt that is trying to adapt the button-type zinc-air battery for use with larger devices like laptops and mobile phones. Since zinc-air batteries have a much higher energy density than the traditional lithium-ion batteries used currently with laptops, ReVolt's ventures could result in a tripling of the battery life of such devices.

Plus, a look at current research hints at the possible replacement of the old-golden alkaline manganese dioxide-based batteries with newer nickel- and lithium-based battery formulas such as nickel oxyhydroxide, olivine-type lithium iron phosphate and nanowires.

Battery technology seems to be a happening area of research, and any significant breakthroughs will provide a good break for the electronics industry.



TOMORROW'S CONNECTED HOME

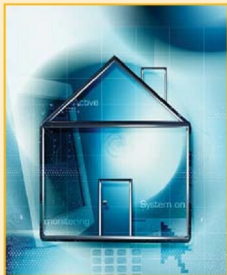
Bill Gates envisioned and implemented a smart and connected home many years ago, but in the next few years even our homes might become connected to a reasonable extent. The term 'connect' is now being used beyond the context of just mobile phones and computers. Numerous devices have the capability to connect to home wireless networks and the Internet – home theatre systems, gaming consoles, digital photo frames, cameras, printers, and more.

Several efforts are on, to take home networking to the next level.

In January 2009, Cisco announced the Cisco Device Connections Program (Cisco DCP), to enable manufacturers to use the Home Network Administration Protocol (HNAP) in a variety of network-connected devices. The company claims that HNAP enables a simple, secure and open approach for devices to be set up and configured with other HNAP-enabled devices on a home network.

On the consumer electronics front, news reports say that companies like Samsung, Sony and Panasonic are relying on the connected home concept for their next major surge up the ladder, and they already have numerous network-enabled products.

HP offers a range of MediaSmart products including receivers, servers,



software and services that help you consolidate all your media files and share them across a range of connected devices at home. The HP Media Smart Connect, for example, is an advanced digital media receiver that turns any high-definition television (HDTV) into a next-generation TV. You can view all your photos, music, movies and videos on your television by connecting it to multiple computers around the home.

You may dream of being able to open all your media on all your devices. But would this be possible? Would all the formats be compatible with all the devices? To overcome this issue, the Digital Living Network Alliance (DLNA) is working to provide a standard. DLNA-certified products ensure that digital content (photos, video and music) is playable across products

sharing a wired or wireless network connection. Over the next few years, we can expect to see a range of DLNA-certified products in the market.

The other context of home networking is a little more hi-fi. That is, the computerisation of various household features such as temperature control, lighting, etc. and its centralised control through a panel located either inside the house, outside, or through a mobile device. In India, this will probably be more popular in hotels and offices than in homes – at least in the near future.